Yash Lalwani

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Education

Rutgers University–New Brunswick — New Brunswick, NJ
Bachelor of Science in Computer Science, Minor in Data Science — *Expected May 2027*

- **GPA:** 3.85/4.00 | **Honors & Awards:** 1st Place HackRU (Fall 2026), Dean's List (4×)
- **Relevant Coursework:** Deep Learning and Artificial Intelligence, Machine Learning Principles, Object-Oriented Programming and Software Design, Data Structures and Algorithms, Applied Data Science, Python Programming

Work Experience

Research Assistant – AI & Data Automation — Oct 2025 – Present
Rutgers University – Bloustein School — *New Brunswick, NJ*

- Built Python and pandas pipelines to automate quarterly reporting for a transportation workforce training program, consolidating enrollment, financial, and survey data into PDF reports across 8 configurable sections.
- Automated quarterly financial reconciliation across Oracle exports and Microsoft Access records, processing 1,200+ expenditure rows and 390+ log entries in a quarterly run and generating a 13-sheet Excel audit workbook.
- Developed automated training analytics and validation workflows using course-title matching, budget comparisons, and statistical anomaly detection to flag inconsistencies for review.
- Evaluated 5,600+ LLM prompt-response pairs using NLP techniques and Hugging Face APIs, identifying model-performance trends and assessing output quality.

AI Document Intelligence Extern — Sep 2026 – Present
Pfizer Extern — *Remote*

- Developing OCR and RAG pipelines using Tesseract and PaddleOCR to extract and classify pharmaceutical documents.
- Building RAG-based retrieval using LlamaIndex, Mistral, and Phi-2, optimizing document chunking and search.
- Evaluating OCR accuracy, RAG retrieval quality, and routing performance, benchmarking model trade-offs and preparing deployment recommendations through a technical report and interactive demo UI.

TrialWright – 1st Place at HackRU Hackathon — Oct 2026
Regeneron Track — *New Brunswick, NJ*

- Built a Python/FastAPI protocol review engine combining deterministic rules and four specialized LLM reviewers, analyzing a 134-page clinical trial protocol in 31 seconds for approximately $0.01.
- Implemented page-level quote verification and reproducible benchmarks covering 16 error types; achieved 9× the precision of a baseline LLM prompt with zero hallucinated citations in evaluation.

Projects

RAG-Engine | *Python, LangGraph, Qdrant, PostgreSQL, Redis* — Live Demo

- Built an MCP-based RAG engine for document and SQL question answering, combining dense and BM25 retrieval with Reciprocal Rank Fusion and cross-encoder reranking across 197 documents, including 150 distractors.
- Implemented advanced retrieval strategies including HyDE, CRAG, and Self-RAG, with citation verification, human-approved Text-to-SQL, and Redis caching.
- Benchmarked 8 retrieval configurations on 40 questions, achieving 0.82 Recall@5 and 0.94 Ragas faithfulness.

Astra | *Python, FastAPI, LangGraph, PostgreSQL, LangMem* — Live Demo

- Built a multi-agent clinical trial auditing system using 6 parallel LangGraph specialist agents, with an evidence pipeline ingesting 788 registered trials and 758 PubMed papers.
- Designed a three-layer memory architecture using pgvector, SQL sponsor profiles, and LangMem feedback rules; implemented AI guardrails, citation verification, and confidence-gated human review.
- Integrated LangSmith tracing, SSE streaming, and persistent audit replay into a FastAPI backend deployed on Railway.

Plane | *Node.js, TypeScript, Express.js, PostgreSQL, Redis, BullMQ* — Live Demo

- Built a 47-endpoint multi-tenant project management REST API using Express.js and PostgreSQL/Prisma, with project-scoped role-based access control (RBAC) and safeguards against cross-project access.
- Implemented JWT authentication with hashed refresh-token rotation, email verification, and password recovery.
- Developed BullMQ/Redis email workers with retries and backoff, dashboard caching, and database fallback during Redis outages.

Technical Skills

Languages: Python, JavaScript, TypeScript, Java, C, SQL
Frameworks & Libraries: FastAPI, Node.js, Express.js, React, Next.js, pandas, NumPy, scikit-learn, PyTorch
AI Engineering: Retrieval-Augmented Generation (RAG), Tool Calling, Multi-Agent Systems, Model Context Protocol (MCP), Human-in-the-Loop (HITL), LangChain, LangGraph, LangMem, LangSmith, AI Guardrails
Retrieval & Search: BM25, Hybrid Search, Reranking, HyDE, CRAG, Self-RAG, Text2SQL, Citation Verification
Databases & Infrastructure: PostgreSQL, MongoDB, MySQL, Redis, Qdrant, pgvector, BullMQ, Google Cloud Storage
Developer Tools & Deployment: Git, Postman, Docker, GitHub Actions, AWS, Amazon SageMaker